OJIMMA CHIMA PETER

Backend & Full-Stack Engineer | Distributed Systems | AI Platforms

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PROFESSIONAL SUMMARY

Backend Engineer with 4+ years of experience architecting production-grade backend systems, distributed services, and AI-powered applications across fintech, SaaS, and developer tooling. Built and supported highly performant systems across multi-tenant platforms, payment systems, real-time services, and event-driven architectures. Strong with Python, TypeScript, Node.js, PostgreSQL, Redis, AWS, and asynchronous processing, with hands-on experience orchestrating multi-provider LLMs, RAG pipelines, and agentic workflows involving 6+ tools and up to 64 concurrent AI agents. Experienced with CI/CD, Docker, cloud infrastructure, and production observability, owning systems end-to-end from architecture and API development through deployment and production operations.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, Go

Backend: FastAPI, NestJS, Node.js, Express.js, Gin, REST, Websockets, SSE

Architecture: Microservices, CQRS, Event-Driven (Blinker, SignalR), Clean / Hexagonal, DDD, SOLID, Modular Monolith

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Neo4j

Cloud & DevOps: AWS (EC2, S3, ECS, RDS), GCP(Cloud Run, Cloud SQL, GCE, Cloud Build), Azure, Docker, Pulumi, GitHub Actions, GitLab CI/CD

AI / ML: LangChain, LangGraph, RAG, LangSmith, CrewAI, multi-provider LLM orchestration (Open AI, Gemini, Groq), JSON Patch editing

Observability: Sentry, OpenTelemetry, Prometheus, structured logging, correlation IDs, Arize Phoenix(tracing)

Testing & Quality: Pytest, Jest, Vitest, Integration Tests, OpenAPI/Swagger, ADR Documentation, ESLint, Prettier

PROFESSIONAL EXPERIENCE

Backend Engineer — easyspecs.ai **March 2026 – Present | Spain**

- Co-built the EasySpecs headless CLI (TypeScript, Node.js 18+, Commander.js), orchestrating AI agent prompts through a Factory → Pipeline → Workstation architecture with six pipelines: synthesis, coverage, remediation, link-mapping, upload, and download.
- Designed an adaptive concurrency pool with duration-based pressure control (step-down/step-up), dynamically throttling parallel AI workstation execution based on completion times and configurable thresholds • Implemented a pluggable agent harness system supporting multiple runtimes (OpenCode, Claude Code, Gemini CLI, Cursor CLI, Codex) with unified spawn adapters and optional remote worker execution via REST API.
- Built git worktree isolation for analysis runs with JSON snapshot persistence, enabling resume-on-interrupt and automatic stale worktree recovery via git worktree prune • Co-architected the context drift pipeline comparing generated documentation against reference specs, with zero-reference remediation pools that spawn AI agents to document undocumented code paths.

- Implemented JSON schema validation (ajv, ajv-formats) with automatic malformed LLM output recovery using jsonrepair, supporting coordination list repair attempts and markdown evidence validation.
- Designed the configuration layer with schema-versioned config.json, layered merge precedence (file → CLI flags), provider API key injection, and sensitive value redaction for diagnostics output.
- Built observability tooling: append-only run logs, live Markdown performance reports with per-workstation timing, pool pressure event tables, and structured JSON stdout for CI parsing.
- Implemented bidirectional cloud sync with EasySpecs REST API (batch upload/download, auth login/logout, project ID binding) with optional cloud context cache to skip redundant analysis runs.

Tech Stack: *TypeScript, Node.js, Commander.js, esbuild, AJV, jsonrepair, Git Worktrees, REST APIs, Stripe*

Fullstack Developer — Gluecharm **October 2025 – March 2026 | Spain**

- Architected an event-driven microservices platform for AI-powered content generation serving multi-tenant workloads, integrating Neo4j (graph relations + versioned content nodes) and PostgreSQL in a dual-persistence design each store selected for its query strengths, not convenience.
- Designed a vendor MIME-type API versioning strategy (application/vnd.gluecharm. v1+json) enabling backward-compatible contract evolution across client versions without URL proliferation. • Built a schema-driven artifact system where JSON schema definitions govern content types, field relationships, and AI-editable surface area new content types added without modifying core CRUD logic.
- Implemented JSON Patch (RFC 6902) based LLM editing: instead of full-document regeneration, the AI outputs minimal, schema-validated change operations (add / replace / remove), cutting token usage by ~60% and eliminating data-loss risk on partial updates.
- Provisioned and managed AWS infrastructure (ECS, Lambda, S3) using Pulumi IaC; GitLab CI/CD pipelines (generate → trigger → upload pattern) enabled reproducible deployments across dev, staging, and production with zero manual steps.
- Integrated LangChain / LangGraph for multi-provider LLM orchestration with intelligent model routing by complexity tier, rate limiting, and optional Arize Phoenix tracing for AI observability in production.

Tech Stack: *Python, FastAPI, Neo4j, PostgreSQL, Redis, Celery, LangChain, LangGraph, OpenAI, Gemini, Mistral, Groq, OpenRouter, Vapi, Vue 3, TypeScript, Vite, Tailwind CSS, Pinia, Vitest, Playwright, Supabase, AWS ECS/Fargate/Lambda/S3, Pulumi, Docker, GitLab CI/CD*

Backend & DevOps Engineer — Manom **April 2023 – September 2025 | Africa**

- Engineered a modular NestJS/TypeScript fintech backend with 246 API endpoints across 31 modules, supporting invoicing, accounting, payroll, subscriptions, CRM, wallets, and analytics.
- Built Redis- and Bull-powered background processing for invoice generation, payment provisioning, subscription charging, retries, reminders, and grace-period management.
- Integrated Paystack, NOWPayments, and Finecore APIs for fiat and cryptocurrency payments, virtual accounts, wallets, bank transfers, foreign exchange, and identity verification.
- Developed end-to-end invoice workflows supporting partial payments, fiat/crypto checkout, PDF generation, cloud storage, scheduled delivery, and overdue notifications.
- Built a Gemini-powered financial assistant with LangChain tool calling, routing conversations across six tools for invoice generation, business onboarding, subscriptions, FAQs, and financial-data retrieval, backed by FastAPI and MongoDB chat history.

- Delivered production infrastructure with real-time SSE notifications, structured Winston logging, readiness checks, Docker Swarm, Nginx ingress, Redis, automated TLS, rolling deployments, rollback protection, and GitHub Actions CI/CD on Azure VPS.

Tech Stack: *TypeScript, NestJS, Prisma, PostgreSQL, Redis, Bull), Node.js, Express, Cloudinary, Nodemailer, Puppeteer, PDFKit, Handlebars, OpenAPI/Swagger, EventEmitter2, SSE, GitHub Actions CI/CD, Docker Swarm · Python, FastAPI, Langgraph, Gemini, MongoDB*

Frontend Developer — BuildWithSeamless **August 2022 – March 2023 | Nigeria**

- Designed and developed responsive, production-ready web applications for clients across education, media, logistics, real estate, and fintech, translating business requirements into functional user experiences.
- Built an **AI-powered study platform** featuring interactive learning workflows and AI-assisted study experiences, alongside a podcast platform with content discovery and media consumption interfaces.
- Developed frontend experiences for a logistics management platform and a US-based real estate agency, implementing intuitive dashboards, data-driven interfaces, and streamlined business workflows.
- Built and marketed a **real-time funding divergence screener** for ContextProtocol, developing interfaces for monitoring and analyzing funding-rate discrepancies across cryptocurrency markets.
- Integrated REST APIs and third-party services into frontend applications, implementing authentication flows, asynchronous data handling, error management, and reusable API integration patterns.
- Established reusable UI components and responsive design patterns across projects, leveraging modern frontend tooling to improve maintainability, development efficiency, and cross-device consistency.

Tech Stack*: React, TypeScript, JavaScript, Next.js, Tailwind CSS, REST APIs, Axios, React Hook Form, Zod, WebSockets, Git, GitHub.*

SELECTED PROJECTS

AI Manager API – GlueX Platform

- Built and owns the AI layer for the GlueX platform: multi-provider LLM orchestration (OpenAI, Gemini, Groq, Mistral) via a modular monolith with 8 domain modules, each with router/service/schema separation and DI wiring via dependency-injector containers.
- Designed a two-phase semantic edit flow: (1) direct edit — user intent → embedding-based semantic search → artifact selection → JSON Patch generation; (2) propagated edit — post-change semantic search finds related artifacts and generates consistency patches with confidence scores. Pre-LLM narrowing via type filtering + vector search reduces prompt cost significantly.
- Shipped streaming SSE conversational agents and VAPI voice-agent integration for multimodal interaction; artifact JSON schemas stored in S3 as the authoritative source of LLM-editable surface area.
- Maintained with spec-driven development (OpenSpec change proposals with ADDED/MODIFIED/REMOVED requirement deltas and scenarios), validated before implementation and archived post-deployment.

Tech Stack: *Python 3.12, FastAPI, LangChain, LangGraph, DSPy, OpenAI, Gemini, Groq, Mistral, Celery, Redis, AWS S3, Pydantic v2, dependency-injector, Blinker, Sentry, Arize Phoenix, pytest, Ruff, Pulumi, GitLab CI/CD, Docker*

StudyBuddy — AI-Powered Study Companion

- Built a FastAPI-based study companion that lets students upload materials, ingest them into a hybrid search index, tutor via grounded RAG chat, and auto-generate chapter study cards with MCQ quizzes, covering an async parse → chunk → embed → index pipeline, and LangGraph agents over WebSocket.
- Built long-term learner memory management to extract and retrieve evolving personal, academic, learning, and progress context over asynchronous background operations with resilient RabbitMQ workers having retries and DLQs, enabling personalized answers across conversations and sessions.
- Added LangSmith evaluations for RAG and agent workflows, measuring retrieval relevance, response groundedness, and answer quality across test cases.

Tech Stack: *Python 3.14+, FastAPI, Supabase Postgres/Storage, Redis, RabbitMQ (aio-pika), Elasticsearch 8 (hybrid RRF/kNN/BM25), Sentence Transformers (all-MiniLM-L6-v2), LangGraph, Google Gemini, Unstructured, Docker*

ADI Camping — Multi-Tenant Camping Property Management Platform

- Built a full-stack multi-tenant camping property management and booking platform covering staff operations and public guest flows: reservations, seasonal availability, check-in/out, billing, rentals, shop management, campsite maps, and Spanish tourism compliance (police reports, tourist tax) with multi-location tenancy via reactive Pinia stores.
- Integrated Stripe payments, Twilio messaging, AI guest-document OCR via Gemini (no image retention), Vertex AI EU RAG for staff/guest chat, a WordPress Gutenberg/shortcode embed plugin for guest booking, json-rules-engine for pricing/cancellation/rentals, and AWS S3 + CloudFront static delivery via Pulumi IaC.

Tech Stack: *Vue 3, Tailwind CSS, Supabase (Postgres, Auth, RLS, Edge Functions/Deno), Stripe, Twilio, Google Gemini, Vertex AI (EU RAG), Vapi, Playwright, Pulumi, AWS S3/CloudFront, WordPress (PHP Gutenberg block + shortcode), json-rules-engine*

EDUCATION

Bachelor of Science in Computer Science **2022 - 2026**

University of Benin - (Upper Class Honours)